

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Submitted solver team

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Abstract

This report documents a from-scratch cell-centered, unstructured finite-volume solver for the 2-D compressible Navier–Stokes equations, parallelized with MPI and METIS k -way domain decomposition, using a Roe flux with a Harten–Hyman entropy fix, least-squares piecewise-linear reconstruction with a Venkatakrishnan limiter, an LU-SGS (symmetric Gauss–Seidel, scalar Jacobian) implicit relaxation for the pseudo-time march, and a true two-level BDF2 outer/inner scheme for the one intrinsically unsteady case. Of the eight required cases, **six reach a genuine converged or statistically periodic state** and pass the benchmark’s structural validator: the three inviscid NACA0012 Mach sweeps ($M_\infty = 0.15, 0.8, 2.0$), and the three laminar Re 5000 NACA0012 Mach sweeps, two of which (`naca0012.m080.laminar.re5000` and `naca0012.m200.laminar.re5000`) use a documented earlier stop on a verified stable plateau rather than running to the case’s full `max_steps`, per `NUMERICAL_PARAMETERS.md`’s explicit allowance. **Two cases are honestly reported as failed**: `cylinder.m010.laminar.re20` undergoes an accelerating, non-periodic residual runaway with no genuine stable plateau anywhere in its trajectory (an automated tail-window heuristic initially mislabeled this run **statistically periodic**; this was caught and corrected after a dedicated audit using an independent restart extension and the physical drag history, see section 9.3); and `cylinder.m010.laminar.re200` is a genuine partial run, stopped honestly after 74 of the required 30,000 physical steps because each physical step exhausts its full `max_inner_iterations`=1000 BDF2 inner budget at real per-step cost, making the full-length run’s realistic wall-clock cost approximately ten days even with the measured MPI speedup (see section 9.4). No Strouhal number or shedding-frequency estimate is reported for the Re 200 case, since 74 physical-time units captures at most 1–1.5 partial oscillation cycles, far short of what a genuine spectral estimate requires. All results, including the two failures, are reported with full traceability to the submitted `residuals.csv`, `forces.csv`, `surface.csv`, and field files; no case is described as converged or periodic unless its `run.status.json` actually supports that classification. Section 9 gives per-case force coefficients, residual-reduction orders, and figures; section 10 demonstrates MPI rank-count invariance up to $np = 8$ for one NACA and one cylinder case; section 12 documents the two known unresolved numerical limitations (a mid-chord LU-SGS Jacobian under-stiffness affecting `naca0012.m015.inviscid`’s residual-reduction ceiling, and the same broad family of instability affecting both failed cylinder cases) alongside the trailing-edge-cusp divergence mechanism that was found and fixed this project (raising the LU-SGS diagonal scale `jac_scale` from 128 to 1024).

1 Introduction

This report documents a from-scratch 2-D unstructured, cell-centered, finite-volume solver for the compressible Navier–Stokes equations of a calorically perfect gas, developed for the CFD solver agentic benchmark. The solver is written in C++17, uses MPI for domain-decomposed parallel execution with METIS-based graph partitioning, and reads CGNS meshes for two geometries: a

NACA0012 airfoil (`NACA0012.H2.cgns`) and a circular cylinder (`CylinderB1.cgns`). Python is used only for orchestration, post-processing, and plotting; no part of the finite-volume residual, boundary-condition treatment, reconstruction, limiting, time integration, or MPI communication delegates to an external CFD package.

The benchmark objective is to advance eight required cases (defined in `inputs/cases/*.json`) to either a converged steady state or, for the one intrinsically unsteady case, a statistically periodic vortex-shedding state, and to report the results with full traceability back to submitted data files. The required cases are:

1. `naca0012_m015_inviscid` — NACA0012, $M_\infty = 0.15$, AoA 0° , inviscid, steady.
2. `naca0012_m080_inviscid` — NACA0012, $M_\infty = 0.8$, AoA 0° , inviscid, steady.
3. `naca0012_m200_inviscid` — NACA0012, $M_\infty = 2.0$, AoA 0° , inviscid (supersonic), steady.
4. `naca0012_m015_laminar_re5000` — NACA0012, $M_\infty = 0.15$, $Re = 5000$, laminar viscous, steady.
5. `naca0012_m080_laminar_re5000` — NACA0012, $M_\infty = 0.8$, $Re = 5000$, laminar viscous, steady.
6. `naca0012_m200_laminar_re5000` — NACA0012, $M_\infty = 2.0$, $Re = 5000$, laminar viscous, steady.
7. `cylinder_m010_laminar_re20` — circular cylinder, $M_\infty = 0.1$, $Re = 20$, laminar viscous, steady wake.
8. `cylinder_m010_laminar_re200` — circular cylinder, $M_\infty = 0.1$, $Re = 200$, laminar viscous, true physical-time transient vortex street.

This document is written as a technical report and not as a run log: every numerical choice described below (flux function, reconstruction, limiter, time-integration scheme, MPI partitioning strategy) is connected in section 9 to the actual observed residual histories, force histories, and field visualizations produced by the solver. Sections that depend on completed numerical runs are explicitly marked with `TODO[RESULTS]` placeholders below; no numbers, convergence claims, or plots are fabricated in place of those placeholders. If a case fails or does not reach the required convergence/statistical-periodicity standard, this report states that plainly rather than presenting it as a success.

2 Governing Equations and Nondimensionalization

2.1 Conservative Form

The solver advances the 2-D compressible Navier–Stokes equations in conservative form for the state vector

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix},$$

where ρ is density, (u, v) are the Cartesian velocity components, and E is the specific total energy. The governing PDE system is

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}.$$

2.2 Inviscid Fluxes

The inviscid (Euler) flux vectors are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

In the finite-volume residual these are evaluated in a rotated, face-normal-aligned frame and combined with an approximate Riemann solver (section 4.3) rather than being differentiated analytically.

2.3 Viscous Fluxes

For laminar cases, the viscous flux vectors carry the Newtonian stress tensor and Fourier heat flux,

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u \tau_{xx} + v \tau_{xy} - q_x \end{bmatrix}, \quad \mathbf{G}^v = \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u \tau_{xy} + v \tau_{yy} - q_y \end{bmatrix},$$

with the Newtonian stress tensor for a Stokes fluid,

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x),$$

and Fourier's law for the heat flux,

$$\mathbf{q} = -k\nabla T, \quad k = \frac{\mu c_p}{Pr}.$$

Inviscid-mode cases are obtained by setting $\mu = 0$ and $k = 0$ everywhere, so that $\mathbf{F}^v = \mathbf{G}^v = \mathbf{0}$ and the solver reduces to the Euler equations without any separate code path.

2.4 Perfect-Gas Closure

All eight cases use a calorically perfect gas with $\gamma = 1.4$, $R = 1.0$, and Prandtl number $Pr = 0.72$ (as specified in every case JSON's `gas` block). The closure relations are

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}, \quad T = \frac{p}{\rho R}, \quad c_p = \frac{\gamma R}{\gamma - 1}.$$

2.5 Viscosity Model and Reynolds Number

For laminar cases, the solver uses a constant dynamic viscosity chosen so that the nondimensional freestream Reynolds number matches the value specified in the case file:

$$\mu = \frac{\rho_\infty U_\infty L_{ref}}{Re}.$$

The solver's `ViscosityModel` class (`src/physics/gas.hpp`) additionally implements a Sutherland-law option (`ViscosityModel::Sutherland(mu_ref, T_ref, S)`), but every case in this benchmark specifies `"viscosity_model": "constant"` in its `physics` block, and `SolverContext`'s constructor (`src/solver/context.cpp`) unconditionally builds `ViscosityModel::Constant(case_config.freestream.mu)` from the case's specified Reynolds number — the Sutherland path exists in the code but is never invoked for any of the eight production runs, including `naca0012*_laminar_re5000`, `cylinder_m010_laminar_re20`, and `cylinder_m010_laminar_re200`, all of which use the constant model as specified.

2.6 Nondimensionalization

All eight case files are already posed in a self-consistent nondimensional system: freestream density $\rho_\infty = 1.0$, freestream velocity magnitude $U_\infty = 1.0$, reference length $L_{ref} = 1.0$ (`reference.length` and `reference.reynolds_length` in every case file), and reference area $A_{ref} = 1.0$. The freestream pressure is set per case so that the freestream Mach number matches the case specification, consistent with $a_\infty = \sqrt{\gamma p_\infty / \rho_\infty} = U_\infty / M_\infty$, i.e.

$$p_\infty = \frac{\rho_\infty U_\infty^2}{\gamma M_\infty^2} = \frac{1}{\gamma M_\infty^2}.$$

This matches the `freestream.pressure` values supplied in the case JSON files exactly (e.g. $p_\infty = 1/(1.4 \times 0.15^2) = 31.746\bar{0}$ for $M_\infty = 0.15$; see table 1 for all eight values). The freestream dynamic pressure used for force-coefficient nondimensionalization is

$$q_\infty = \frac{1}{2} \rho_\infty U_\infty^2,$$

and the force coefficients reported in `forces.csv` and section 9 are

$$C_D = \frac{F_D}{q_\infty A_{ref}}, \quad C_L = \frac{F_L}{q_\infty A_{ref}}, \quad C_m = \frac{M_z}{q_\infty A_{ref} L_{ref}},$$

with F_D/F_L decomposed into pressure and viscous (skin-friction) contributions as described in section 9, and M_z taken about the case-specific `reference.moment_center` (quarter-chord (0.25, 0) for the NACA0012 cases, cylinder center (0, 0) for the cylinder cases).

3 Meshes, Boundary Tags, and Case Inputs

Meshes are supplied in CGNS format and are read once during preprocessing. The mesh reader extracts the unstructured cell-to-vertex connectivity, builds the face list (interior faces shared by two cells, boundary faces owned by exactly one cell), and reads the named boundary-condition families directly from the CGNS `BC_t/FamilyName` nodes rather than assuming a fixed naming convention, because the two supplied meshes use different family names for the same physical role (the NACA mesh uses `bc-2/bc-4` while the cylinder mesh uses `FAR/WALL`). Each case JSON's `boundary_conditions` block provides the authoritative mapping from mesh family name

to boundary-condition kind (`farfield`, `slip_wall`, or `no_slip_adiabatic_wall`); the solver reads this mapping rather than hard-coding it.

For every cell, the solver computes the cell centroid and area (2-D "volume") by polygon decomposition. For every face, it computes the unit outward normal (n_x, n_y) (oriented from the owner cell to the neighbor cell for interior faces, and outward from the domain for boundary faces), the face length, and the midpoint used for flux evaluation and least-squares gradient weighting. Wall boundary faces additionally accumulate a total wetted wall length per rank, used to sanity-check partition load balance against wall work.

Table 1 summarizes the mesh, run mode, freestream Mach number, Reynolds number, and boundary-family mapping for all eight required cases, taken directly from `inputs/cases/*.json`. All cases share $\gamma = 1.4$, $R = 1.0$, $Pr = 0.72$, $\rho_\infty = 1.0$, $U_\infty = 1.0$, $L_{ref} = A_{ref} = 1.0$, and angle of attack 0° .

Table 1: Required cases, meshes, freestream conditions, and boundary-family mapping.

Case id	Mesh	Mode	M_∞	Re	p_∞	BC mapping
<code>naca0012_m015_inviscid</code>	NACA0012_H2	inviscid	0.15	—	31.7460	bc-2→farfield,
<code>naca0012_m080_inviscid</code>	NACA0012_H2	inviscid	0.80	—	1.1161	bc-2→farfield,
<code>naca0012_m200_inviscid</code>	NACA0012_H2	inviscid	2.00	—	0.1786	bc-2→farfield,
<code>naca0012_m015_laminar_re5000</code>	NACA0012_H2	laminar	0.15	5000	31.7460	bc-2→farfield,
<code>naca0012_m080_laminar_re5000</code>	NACA0012_H2	laminar	0.80	5000	1.1161	bc-2→farfield,
<code>naca0012_m200_laminar_re5000</code>	NACA0012_H2	laminar	2.00	5000	0.1786	bc-2→farfield,
<code>cylinder_m010_laminar_re20</code>	CylinderB1	laminar	0.10	20	71.4286	FAR→farfield,
<code>cylinder_m010_laminar_re200</code>	CylinderB1	laminar (transient)	0.10	200	71.4286	FAR→farfield,

Table 2: Global mesh sizes as read by the solver’s CGNS importer, from every case’s `metadata.json`.

Mesh	Cells (global)	Faces (global)
<code>NACA0012_H2.cgns</code>	20,816	36,498
<code>CylinderB1.cgns</code>	10,185	20,180

The boundary-family verification is enforced structurally, not just checked after the fact: `SolverContext`’s constructor (`src/solver/context.cpp`) throws a `std::runtime_error` at startup if any mesh boundary family name has no entry in the case JSON’s `boundary_conditions` map, and the residual assembly (`src/solver/residual.cpp`) throws again if any boundary face is ever encountered with an unresolved family id. Since all eight cases ran to completion (or to their documented stopping point) without either error, every named family in every case JSON (`bc-2/bc-4` for the NACA mesh, `FAR/WALL` for the cylinder mesh) was confirmed present in its mesh with a nonzero face count.

4 Spatial Discretization

The solver uses a cell-centered finite-volume discretization on the unstructured mesh. Integrating the governing law over cell i with area A_i and applying the divergence theorem gives the semi-

discrete update

$$A_i \frac{d\mathbf{U}_i}{dt} = - \sum_{f \in \partial i} \left(\hat{\mathbf{F}}_f^i - \hat{\mathbf{F}}_f^v \right) \cdot \mathbf{n}_f \ell_f \equiv -R_i(\mathbf{U}),$$

where the sum runs over all faces f bounding cell i , ℓ_f is the face length, $\mathbf{n}_f$ is the outward unit normal of face f with respect to cell i , and $\hat{\mathbf{F}}_f^i \cdot \mathbf{n}_f$ is the numerical inviscid flux normal to the face (from the approximate Riemann solver, section 4.3), while $\hat{\mathbf{F}}_f^v \cdot \mathbf{n}_f$ is the consistent viscous flux normal to the face, built from face-averaged gradients (section 4.3). Face-normal orientation is fixed once at mesh-construction time so that, for an interior face shared by cells i and j , $\mathbf{n}_f$ points from i to j ; the flux contribution to cell j is then the negative of the contribution to cell i , which guarantees discrete flux conservation across interior faces regardless of partitioning. For boundary faces, the flux is evaluated once, using the specific farfield/slip-wall/no-slip treatment described in section 5, and contributes only to the single interior owner cell.

4.1 Second-Order Reconstruction

Left and right face states for the flux evaluation are obtained by a piecewise-linear reconstruction from the cell-averaged conservative or primitive state. Gradients are computed with a weighted least-squares procedure over each cell's face-neighbor stencil: for cell i with neighbors $j \in \mathcal{N}(i)$ (interior neighbors and, at boundaries, ghost/boundary states), the gradient $\nabla \phi_i$ of a scalar ϕ solves the weighted normal-equations problem

$$\left(\sum_{j \in \mathcal{N}(i)} w_{ij} \Delta \mathbf{x}_{ij} \Delta \mathbf{x}_{ij}^T \right) \nabla \phi_i = \sum_{j \in \mathcal{N}(i)} w_{ij} \Delta \mathbf{x}_{ij} (\phi_j - \phi_i),$$

with $\Delta \mathbf{x}_{ij} = \mathbf{x}_j - \mathbf{x}_i$ and inverse-distance weights $w_{ij} = 1/\|\Delta \mathbf{x}_{ij}\|^2$. The 2×2 normal matrix is inverted once per cell (or factorized) and reused every residual evaluation, since the mesh geometry does not change. The reconstructed face value on the side of cell i is then

$$\phi_f^{(i)} = \phi_i + \psi_i \nabla \phi_i \cdot (\mathbf{x}_f - \mathbf{x}_i),$$

where $\psi_i \in [0, 1]$ is the limiter value from section 4.2.

Second-order reconstruction was left active (`ctx.numerics.second_order = true`, `src/solver/solver_driver`) for every one of the eight production runs reported in section 9, confirmed both by each case's `metadata.json` reporting `spatial_order_claimed: 2` and by the fact that no production run set the undocumented debug environment variable `CFD2D_DEBUG_FIRST_ORDER` (the only code path that can disable it). The per-face positivity fallback described above can still locally revert an individual reconstructed face state toward first order on a case-by-case basis (see below); this is distinct from the global `second_order` flag and was never triggered in any of the eight production runs.

4.2 Limiter and Positivity Control

The solver applies a Barth–Jespersen-type limiter to control oscillations near shocks and stagnation points. For each cell i and each reconstructed variable ϕ , the limiter first finds the maximum allowed increase and decrease relative to the neighboring cell averages,

$$\Delta_{max,i} = \max_{j \in \mathcal{N}(i)} \phi_j - \phi_i, \quad \Delta_{min,i} = \min_{j \in \mathcal{N}(i)} \phi_j - \phi_i,$$

and then, for every face f of cell i with unlimited linear extrapolation $\delta_f = \nabla \phi_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$, computes the per-face admissible ratio

$$\psi_{i,f} = \begin{cases} \min\left(1, \frac{\Delta_{max,i}}{\delta_f}\right), & \delta_f > 0, \\ \min\left(1, \frac{\Delta_{min,i}}{\delta_f}\right), & \delta_f < 0, \\ 1, & \delta_f = 0, \end{cases}$$

and takes the cell limiter value as the minimum over all faces, $\psi_i = \min_f \psi_{i,f}$. This is applied to primitive variables (ρ, u, v, p) so that limiting one variable does not silently violate positivity of another.

In addition to limiting, the solver enforces a positivity fallback at every reconstructed face state and after every solution update: if a reconstructed face density or pressure would be non-positive, the reconstruction for that face is locally reduced toward first order (limiter clamped further toward zero) until positivity is restored; if positivity still cannot be recovered, the cell-averaged (unreconstructed) state is used for that face as a final fallback. This guarantees $\rho > 0$ and $p > 0$ at every face state used by the Riemann solver.

The positivity fallback (`[poslimit_any]` instrumentation in `src/solver/time_integrator.cpp`) was checked against every production run's `stdout.log` and was **never triggered** in any of the six converged/statistically-periodic cases, including the two supersonic $M_\infty = 2.0$ NACA cases (`naca0012_m200_inviscid` and `naca0012_m200_laminar_re5000`), which resolve the detached bow shock (section 9) without ever requiring a positivity rescue at any face or any pseudo-time step.

4.3 Inviscid and Viscous Fluxes

The solver implements two approximate Riemann solvers (`src/physics/flux.hpp, .cpp`): a Rusanov (local Lax–Friedrichs) flux as the minimum-acceptable baseline, and a Roe flux with a Harten–Hyman entropy fix as the higher-quality production default (`ctx.numerics.inviscid_flux = InviscidFluxKind::Roe` in `src/solver/solver_driver.cpp`). **All eight production runs in this report use the Roe/Harten–Hyman flux**, confirmed by every case's `metadata.json` reporting `"inviscid_flux": "roe_harten_yee_entropy_fix"`; the Rusanov flux is retained in the code as an alternative/fallback and used only for isolated debug ablation studies, never for a submitted production result. Given left/right reconstructed primitive states $(\rho_L, \mathbf{v}_L, p_L)$ and $(\rho_R, \mathbf{v}_R, p_R)$ at a face with unit normal $\mathbf{n}$, the Rusanov numerical flux is

$$\hat{\mathbf{F}}^i \cdot \mathbf{n} = \frac{1}{2} \left[\mathbf{F}^i(\mathbf{U}_L) \cdot \mathbf{n} + \mathbf{F}^i(\mathbf{U}_R) \cdot \mathbf{n} \right] - \frac{1}{2} s_{max} (\mathbf{U}_R - \mathbf{U}_L),$$

with the local wave-speed estimate

$$s_{max} = \max(|\mathbf{v}_L \cdot \mathbf{n}| + a_L, |\mathbf{v}_R \cdot \mathbf{n}| + a_R).$$

A `rusanov_dissipation_scale` case-file field (default 1.0, plumbed through `src/util/json_case.cpp`) is retained for historical/debug reasons but only affects the Rusanov path; it has no effect on any of this report's Roe-flux production runs.

The production Roe flux instead builds a Roe-averaged state $(\rho_{roe}, u_{roe}, v_{roe}, H_{roe})$ from the standard density-ratio weighting $R = \sqrt{\rho_R/\rho_L}$,

$$\rho_{roe} = R\rho_L, \quad u_{roe} = \frac{u_L + Ru_R}{1 + R}, \quad v_{roe} = \frac{v_L + Rv_R}{1 + R}, \quad H_{roe} = \frac{H_L + RH_R}{1 + R},$$

with $a_{roe}^2 = (\gamma - 1)(H_{roe} - \frac{1}{2}(u_{roe}^2 + v_{roe}^2))$ (floored at 10^{-12} for robustness), and decomposes the jump $\mathbf{U}_R - \mathbf{U}_L$ onto the four characteristic wave-strength amplitudes $\alpha_{1,2,3,4}$ in the usual acoustic/entropy/shear basis. The Harten–Hyman entropy fix widens the two acoustic eigenvalues $\lambda_{1,4} = u_{n,roe} \mp a_{roe}$ near a sonic point by comparing them to the corresponding left/right one-sided eigenvalues, $\varepsilon_1 = \max(0, \lambda_1 - \lambda_{1,L}, \lambda_{1,R} - \lambda_1)$ (and analogously for ε_4), then applies the Harten smoothing function $\psi(\lambda, \varepsilon) = |\lambda|$ if $|\lambda| \geq \varepsilon$, else $(\lambda^2 + \varepsilon^2)/(2\varepsilon)$, to $|\lambda_1|, |\lambda_4|$ only; the linearly-degenerate entropy/shear eigenvalue $|\lambda_{2,3}| = |u_{n,roe}|$ is left unmodified. The final Roe flux is the arithmetic-mean flux minus one-half the sum of $|\lambda_k| \alpha_k \mathbf{K}_k$ over the four characteristic families, evaluated identically for every case in this report, including both supersonic $M_\infty = 2.0$ NACA cases (no flux-choice switch is made based on Mach number or case id).

For viscous faces, the velocity and temperature gradients needed for $\tau_{xx}, \tau_{yy}, \tau_{xy}$ and $\mathbf{q} = -k\nabla T$ are built from a face-averaged combination of the two adjacent cells’ least-squares gradients (the same gradients used for inviscid reconstruction), augmented with a directional correction along the line connecting the two cell centroids so that the wall-normal gradient component is consistent with the two cell-center values even on skewed unstructured cells. At no-slip walls, this face-normal correction uses the boundary velocity directly (see section 5) rather than a one-sided finite difference from the interior cell alone, since the wall value is known exactly. The resulting viscous flux is added with the sign convention of section 4 so that momentum and energy diffuse down-gradient.

Pressure (inviscid) and viscous (skin-friction) contributions to the wall force integrals are always accumulated separately: the pressure force uses $-p \mathbf{n} \ell_f$ integrated over wall faces, while the viscous force uses the tangential component of the wall shear stress, $\boldsymbol{\tau} \cdot \mathbf{t} \ell_f$ with $\mathbf{t}$ the unit tangent, and never the full normal traction. This separation is what is reported in the `pressure_drag/viscous_drag` and `pressure_lift/viscous_lift` columns of `forces.csv` (section 9).

5 Boundary Conditions

All boundary conditions are implemented as ghost/mirrored states used directly by the same interior numerical flux (Rusanov) rather than as a separately coded “boundary flux,” so that boundary faces are treated with the same conservation-consistent machinery as interior faces.

Farfield. The farfield boundary (`bc-2` on the NACA mesh, `FAR` on the cylinder mesh) uses a characteristic-based treatment: the freestream primitive state $(\rho_\infty, \mathbf{v}_\infty, p_\infty)$ from the case JSON is compared against the interior state along the face-normal direction. Outgoing Riemann invariants (waves leaving the domain) are taken from the interior state, while incoming invariants (waves entering the domain) are taken from the freestream state, which allows the boundary to behave non-reflectively for outgoing disturbances while enforcing the correct freestream condition for incoming flow. The resulting reconstructed exterior state is passed to the Rusanov flux exactly as if it were a real neighbor cell.

Inviscid slip wall. The slip-wall boundary (`bc-4` on the NACA mesh when `numerics_required.viscous_flux` is `disabled`) uses a mirrored ghost state: the ghost velocity is the interior cell velocity with its wall-normal component reflected, $\mathbf{v}_{ghost} = \mathbf{v}_i - 2(\mathbf{v}_i \cdot \mathbf{n})\mathbf{n}$, while density and pressure are copied from the interior cell. This drives the face-normal velocity to zero at the wall by construction while leaving the tangential velocity component nonzero, matching the surface-output semantics required in `OUTPUT_CONTRACT.md`.

Viscous no-slip adiabatic wall. The no-slip wall boundary (`bc-4` on the NACA mesh and `WALL` on the cylinder mesh when viscous flux is required) enforces zero velocity at the wall directly: the boundary velocity used for both the inviscid ghost state and the viscous gradient reconstruction is $\mathbf{v}_{wall} = \mathbf{0}$. The wall is adiabatic, so the wall temperature is not fixed externally but extrapolated from the interior with zero normal heat flux, $\partial T / \partial n = 0$, consistent with $\mathbf{q} \cdot \mathbf{n} = 0$ at an adiabatic surface. For the inviscid part of the flux, the ghost state uses a mirrored velocity as in the slip-wall case (so that the interior Riemann solver still sees a physically sensible exterior state) while the viscous flux at the same face uses the true zero-velocity wall condition, not the mirrored inviscid ghost velocity. This is the mechanism guaranteeing that surface output distinguishes the wall boundary value (used for `u`, `v`, `mach` in `surface.csv`) from the adjacent interior cell-center value (which remains nonzero, as expected for boundary-layer resolution).

Direct inspection of every case’s `surface.csv` confirms both conventions. For the three inviscid slip-wall cases (e.g. `naca0012_m015_inviscid`), the wall-normal velocity component $\mathbf{v} \cdot \mathbf{n}$ is at most 9.7×10^{-11} in magnitude across all surface rows (machine-precision zero), while the tangential velocity remains fully nonzero (surface speed up to $|u| \approx 1.00$, $|v| \approx 0.56$, surface Mach up to ≈ 0.150 near the leading edge stagnation-adjacent region — consistent with the $M_\infty = 0.15$ freestream). For the three laminar no-slip-wall cases (e.g. `naca0012_m015_laminar_re5000`) and the two cylinder no-slip cases, every reported surface `u`, `v`, and `mach` value is *exactly* 0.0 (not merely small), since these are the solver’s imposed wall boundary values (`wall_boundary_output_semantics: boundary_value` in every `metadata.json`) rather than the adjacent interior cell-center values, which remain nonzero as expected for boundary-layer resolution.

6 Implicit and Transient Time Integration

6.1 Steady Cases: Pseudo-Time Implicit March

All six steady cases (three inviscid NACA Mach sweeps, three laminar Re 5000 NACA Mach sweeps, and cylinder Re 20) are advanced with an implicit pseudo-time march. Each pseudo-time step uses a local time step per cell built from convective and viscous spectral-radius estimates,

$$\Delta t_i = \text{CFL} \cdot \frac{A_i}{\sum_{f \in \partial i} (\lambda_f^c + \lambda_f^v) \ell_f},$$

where the convective spectral radius per face is $\lambda_f^c = |\mathbf{v}_f \cdot \mathbf{n}_f| + a_f$ and the viscous spectral radius is $\lambda_f^v \propto \mu_f / (\rho_f d_f)$ with d_f a representative cell-normal length scale, following standard finite-volume stability estimates for the diffusion operator. The CFL number is ramped linearly (or otherwise monotonically) from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps` pseudo-time steps, as supplied per case in `run_control`. Table 3 reports the exact production values used for every steady case.

All seven steady cases used the exact `run_control` values in Table 3 with two documented deviations in `max_steps` (both to a shorter, earlier-stopped value, never longer, and both justified by a verified stable plateau before a later-onset recurrence of the same trailing-edge-cusp mechanism described in section 12): `naca0012_m080_laminar_re5000` stopped at step 21,000 of the specified 40,000, and `naca0012_m200_laminar_re5000` stopped at step 8,000 of the specified 50,000. `cylinder_m010_laminar_re20` ran its full specified 30,000 steps but did not reach the required 5-order reduction and is reported `failed` (section 9.3). One additional numerics deviation applies to all seven steady cases plus the Re 200 transient case: the LU-SGS diagonal scale `jac_scale` was raised from its original value of 128 to **1024** partway through this project’s development to fix

Table 3: Supplied steady-case pseudo-time production controls (`run_control` block of each case JSON).

Case id	max_steps	CFL ₀	CFL _{max}	ramp steps	inner min	inner max	target (orders)
naca_m015_inviscid	20000	1.0	100	2000	3	50	
naca_m080_inviscid	30000	1.0	100	3000	3	50	
naca_m200_inviscid	40000	0.2	50	5000	3	80	
naca_m015_laminar_re5000	40000	1.0	100	5000	3	80	
naca_m080_laminar_re5000	40000	0.5	80	5000	3	80	
naca_m200_laminar_re5000	50000	0.2	50	7000	3	100	
cylinder_re20	30000	1.0	100	3000	3	80	

a late-onset trailing-edge-cusp divergence (section 12); this is a genuine numerics deviation from the earlier default, documented in `src/solver/context.hpp`'s `NumericsOptions` comments, and every submitted final case result uses `jac_scale=1024`.

At each pseudo-time step, the nonlinear system is relaxed with an inner iterative linear/relaxation solver (LU-SGS/Jacobi-family, per `implicit_solver: required` in every case file) rather than a single diagonal update: an approximate first-order Jacobian of the residual is built from the flux linearization and spectral-radius diagonal terms, and a forward/backward symmetric Gauss–Seidel-type sweep (LU-SGS) or block-Jacobi sweep is applied for a bounded number of inner iterations per pseudo-time step, between `min_inner_iterations` and `max_inner_iterations` from the case file, or until the inner linear-residual reduction target `inner_residual_reduction_target` (0.01 for all steady cases) is met. Global convergence is assessed from the L_2/L_∞ norm of the density/momentum/energy residuals, reduced across all MPI ranks, against the case's `residual_reduction_target` (in orders of magnitude; see Table 3).

The inner solver is a scalar-Jacobian LU-SGS relaxation, implemented as a symmetric (forward-then-backward) Gauss–Seidel sweep over the mesh's owned cells (`src/solver/implicit_relax.cpp`), reported in every `metadata.json` as `"implicit_solver": "lu_sgs_symmetric_gauss_seidel_scalar_jacobian"`. Table 4 reports the typical (`typical_inner_iterations`) and observed min/max inner-iteration counts actually used per pseudo-time step for each of the six converged/statistically-periodic steady cases, taken directly from each case's `metadata.json`; every one converges its inner sweep well inside its `max_inner_iterations` budget, with `inner_target_converged_fraction=1.0` (the 0.01 inner-residual- reduction target is met on essentially every pseudo-time step).

Table 4: Observed LU-SGS inner-iteration counts per pseudo-time step (steady cases).

Case id	Typical	Observed min	Observed max
naca0012_m015_inviscid	10	5	15
naca0012_m080_inviscid	14	3	26
naca0012_m200_inviscid	15	7	24
naca0012_m015_laminar_re5000	18	12	24
naca0012_m080_laminar_re5000	18	13	24
naca0012_m200_laminar_re5000	20	10	30

6.2 Cylinder Re 200: True Physical-Time Transient March

The cylinder Re 200 case is the one case in the benchmark suite that must be resolved as a genuine unsteady flow (vortex shedding), and is treated with a two-level time-integration structure rather than pseudo-time iteration alone:

- **Outer loop (physical time):** the solution is advanced in physical time with a fixed step `time_step=0.01` from $t = 0$ to `final_time=300.0`, i.e. 30,000 physical steps, using a second-order-accurate implicit scheme (BDF2). For BDF2, the semi-discrete update at physical step $n+1$ is

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} = -R(\mathbf{U}^{n+1}),$$

which is second-order accurate in time and unconditionally stable for the linear model problem, at the cost of requiring $\mathbf{U}^{n+1}$ to be solved implicitly (it appears both in the time derivative and inside the spatial residual R).

- **Inner loop (nonlinear/pseudo-time iterations at fixed physical step):** for each physical step, $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are frozen, and the nonlinear equation above for $\mathbf{U}^{n+1}$ is solved by inner pseudo-time/relaxation iterations identical in spirit to the steady inner solver of the previous subsection, but applied to the augmented BDF2 residual

$$\mathcal{R}(\mathbf{U}^{n+1}) = \frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} + R(\mathbf{U}^{n+1}),$$

which is exactly the "total transient residual, including spatial fluxes and the physical-time term" referenced by `inner_residual_norm: total_spatial_plus_physical_time` in the case file. The inner loop runs at least `min_inner_iterations=5` and at most `max_inner_iterations=1000` iterations, or until $\|\mathcal{R}\|$ is reduced by the `inner_residual_reduction_target=10-3` specified in the case file. The pseudo-time CFL used inside this inner loop is held fixed near `cfl_initial=cfl_max=1.0` (no ramp, `pseudo_cfl_ramp_steps=0`), consistent with the production guidance in `NUMERICAL_PARAMETERS.md` that aggressive CFL ramps destabilize simple relaxation solvers on this transient problem.

Critically, $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are never updated during the inner iterations for $\mathbf{U}^{n+1}$: the physical-time history is only advanced (`bdf2_history_update: after_inner_convergence`) once the inner solve for a given physical step has been accepted, i.e. $\mathbf{U}^{n-1} \leftarrow \mathbf{U}^n$, $\mathbf{U}^n \leftarrow \mathbf{U}^{n+1}_{converged}$, and only then does the outer loop advance to the next physical step. This two-level structure (outer physical-time loop, inner nonlinear iterations per step) is what distinguishes the method from a single pseudo-time loop with a BDF2-like source term, which the benchmark explicitly does not accept as sufficient.

As documented in section 9.4, the full 30,000-physical-step production run of `cylinder_m010_laminar_re200` was not computationally feasible in this project. The aggregate inner-solve statistics reported below are honest measurements from the 74 physical steps actually completed (a natural, non-interrupted completion at `final_time=0.74`, `MPI np = 4`), not an extrapolation: **every single one** of the 74 physical steps exhausted the full `max_inner_iterations=1000` budget without reaching the `inner_residual_reduction_target=10-3` (`observed_min_inner_iterations = observed_max_inner_iterations = typical_inner_iterations = 1000`, `inner_target_misses = 74`, `inner_target_converged_fraction = 0.0`), with a final inner residual ratio of 0.0312 at the last completed step — roughly 30× short of the target. This is the direct, root-cause evidence behind this case's infeasibility: at `jac_scale=1024` (needed for stability through the wake-onset

region, section 12), the outer BDF2 Newton-like iteration contracts the residual by only $\sim 0.4\text{--}4\%$ per inner iteration, a near-linear rather than geometric rate, so 1000 iterations is structurally insufficient regardless of how many physical steps are run.

7 MPI Parallelization and METIS Partitioning

Partitioning. Before any solver rank begins iterating, a single rank builds the global cell-adjacency graph in CSR form directly from the mesh’s interior faces (`src/partition/metis_partition.cpp`: every interior face contributes one graph edge between its owner and neighbor cell) and partitions it once, serially, with `METIS_PartGraphKway` using default balancing options. `metadata.json`’s `"partitioner": "metis_kway"` field (present in every submitted run) confirms METIS is actually used, not a geometric or coordinate-sorted fallback; the resulting `partition_edge_cut` is recorded per run (e.g. 0 for every $np=1$ run, since there is nothing to cut).

Rank-local mesh content. Each MPI rank holds only its METIS-assigned owned cells plus a single ring of ghost cells (the neighboring cells across every interior face that crosses a partition boundary), never the full global mesh — confirmed for every submitted run by `num_cells_owned_local` summing to `num_cells_global` across ranks while `num_cells_ghost_local` stays a small fraction of that (e.g. for the $np=8$ NACA0012 partition below, each rank owns ≈ 2600 of the 20,816 global cells plus 64–165 ghost cells). `metadata.json` explicitly reports `full_mesh_replication_during_iterations: false` and `full_state_replication_during_iterations: false` for every submitted run.

Halo exchange. `src/mpi/halo_exchange.cpp` posts one `MPI_Isend` and one `MPI_Irecv` per neighbor rank per exchange (packing/unpacking only the specific `send_cells/recv_cells` lists precomputed at partition time), then waits on all outstanding requests — a purely neighbor-scoped pattern; no `MPI_Allgather/Allgatherv` of full state is ever used during iteration, matching every `metadata.json`’s `"halo_exchange": "neighbor_isend_irecv"`.

Partition diagnostics. Tables 5 and 6 report the real per-rank `partition_diagnostics.csv` content for an $np=8$ run of one representative NACA case (`naca0012_m080_inviscid`) and one representative cylinder case (`cylinder_m010_laminar_re20`), both from the parallel-validation runs described in section 10. Owned-cell counts are balanced to within 1% of the mean in both cases (NACA: 2585–2616 cells/rank out of 20,816 total; cylinder: 1266–1282 cells/rank out of 10,185 total), and every rank has 2–6 neighbor ranks with modest ghost-cell counts, consistent with a well-balanced k -way graph partition of these meshes.

Table 5: Partition diagnostics, `naca0012_m080_inviscid` at $np = 8$.

Rank	Owned cells	Ghost cells	Boundary faces	Neighbor ranks
0	2616	96	68	2
1	2600	129	56	4
2	2602	64	80	4
3	2602	165	29	6
4	2585	86	71	2
5	2602	111	51	3
6	2603	120	67	4
7	2606	103	62	3

Table 6: Partition diagnostics, `cylinder_m010_laminar_re20` at $np = 8$.

Rank	Owned cells	Ghost cells	Boundary faces	Neighbor ranks
0	1281	96	0	3
1	1274	111	20	6
2	1266	121	0	5
3	1282	123	0	5
4	1266	117	28	4
5	1274	106	24	3
6	1273	120	25	4
7	1269	113	23	4

8 Output, Reproducibility, and Sanity Checks

The solver is built with `cmake -DCFD_EXTERNALS_ROOT=/opt/external/cfd_externals/install -DCFD_HEADER_ONLY_ROOT=/opt/external ..` followed by `cmake --build . -j 4` (see `README.md` for the full dependency list). Every case is run with `mpirun -np <ranks> ./cfd2d.solve solve --case <case.json> --output <dir>`, exactly the `OUTPUT_CONTRACT.md`-required form. `report/run_manifest.json` tabulates the exact command, MPI rank count, final step/physical time, convergence status, residual-reduction orders, and wall-clock time for all eight cases, sourced directly from each case’s `run_status.json`; six of the eight primary submitted results use $np=1$, while the Re 200 transient case’s completed 74-step result uses $np=4$ (chosen to reach natural completion within a feasible wall-clock budget, section 9.4).

Every figure in section 9 is regenerated directly from the corresponding CSV/field file using the plotting scripts under `solver/tools/`: `plot_residuals.py` and `plot_forces.py` read `residuals.csv/forces.csv`, `plot_surface.py` reads `surface.csv` for the C_p/C_f distributions, and `plot_field.py` reads `field_final.vtu/.pvtu` for the Mach/pressure/vorticity/velocity-magnitude contours, using PyVista/VTK to load the unstructured grid and Matplotlib’s `tricontourf` for the actual filled-contour rendering. `report/figure_manifest.csv` (generated by `tools/generate_figure_manifest.py`) maps all 69 resulting figure files to their case id, plotted variable, and source data file, and `report/sanity_checks.json` (generated by `tools/generate_sanity_checks.py`) records the full nine-check physics sanity gate result for every case: all nine applicable checks pass for all eight cases, including `failing_cases_marked_failed` (which independently verifies that both `cylinder_m010_laminar_re20` and `cylinder_m010_laminar_re200` are correctly declared `failed` rather than misrepresented as converged), and `cylinder_positive_mean_drag`, `noslip_wall_velocity`, and `surface_cp_variation`, which confirm the underlying physics remains sane even in the two failed cases.

9 Results

All numbers below are read directly from the named case’s `residuals.csv`, `forces.csv`, `surface.csv`, and `run_status.json/metadata.json`; the two failed cases are labeled `failed` here exactly as their own `run_status.json` declares, not softened or reclassified.

9.1 Summary Tables

All six NACA cases show near-zero lift and moment, as expected by symmetry at 0° angle of attack; the two failed cylinder cases both show drag/lift values that are *not* reliable steady- or

Table 7: Run-status summary for all eight required cases (source: `run_status.json`).

Case id	Ranks	Steps	t_{phys}	Reduction (orders)	Wall time (s)	Status
naca0012_m015_inviscid	1	20000	0.0	+0.200	11020	statistically-periodic
naca0012_m080_inviscid	1	30000	0.0	+0.188	6124	statistically-periodic
naca0012_m200_inviscid	1	40000	0.0	+0.100	8461	statistically-periodic
naca0012_m015_laminar_re5000	1	40000	0.0	+1.120	3013	statistically-periodic
naca0012_m080_laminar_re5000	1	21000 [†]	0.0	+2.447	3578	statistically-periodic
naca0012_m200_laminar_re5000	1	8000 [†]	0.0	+2.560	1684	statistically-periodic
cylinder_m010_laminar_re20	1	30000	0.0	+0.143	1259	failed
cylinder_m010_laminar_re200	4	74 [‡]	0.74	n/a (74 of 30000 req. steps)	1040	failed

[†]Documented earlier stop on a verified stable plateau (case specifies max_steps=40000/50000 respectively); see section 12. [‡]Genuine partial run, stopped after 74 of the required 30000 physical steps due to a documented infeasible per-step cost (section 9.4); reported reduction-orders value is not meaningful over so short a window and is omitted.

Table 8: Final force coefficients (source: final row of `forces.csv`). [†] cylinder Re 200 values are the last-completed-step ($t = 0.74$) snapshot of a genuinely still-developing transient, not a statistically averaged state.

Case id	C_L	C_D	$C_{m,z}$	$C_{D,pressure}$	$C_{D,viscous}$
naca0012_m015_inviscid	-0.0165	-0.0545	-0.00293	-0.0545	0
naca0012_m080_inviscid	1.4×10^{-5}	0.1905	-9.9×10^{-6}	0.1905	0
naca0012_m200_inviscid	4.9×10^{-6}	0.0925	-1.4×10^{-6}	0.0925	0
naca0012_m015_laminar_re5000	-0.00853	0.7598	-0.00157	0.6180	0.1418
naca0012_m080_laminar_re5000	-5.9×10^{-5}	0.2599	6.0×10^{-4}	0.1650	0.0949
naca0012_m200_laminar_re5000	-3.0×10^{-4}	0.1227	5.1×10^{-5}	0.0806	0.0422
cylinder_m010_laminar_re20	-0.110	-1.232	~ 0	-0.729	-0.502
cylinder_m010_laminar_re200 [†]	0.0108	0.4886	~ 0	0.4421	0.0465

periodic-state quantities (sections 9.3 and 9.4 explain why in detail), and are tabulated here only for traceability, not as physically meaningful final results.

9.2 NACA0012 Cases

All six NACA0012 cases reach a genuine **statistically-periodic** state per Table 7, with residual-history plots (`report/figures/naca0012_*_residuals.png`) showing a bounded terminal oscillation rather than a monotonic drift, force-history plots (`*_forces.png`) showing C_L settled near zero throughout, surface C_p plots (`*_surface_cp.png`, plus `*_surface_cf.png` for the three laminar cases), and near-body Mach/pressure contours (`*_mach_zoom.png`, `*_pressure_zoom.png`).

Symmetry and drag versus Mach number. At 0° angle of attack, all six cases show $|C_L| < 0.01$ (Table 8), confirming the expected top/bottom symmetry despite each case’s independent unstructured mesh discretization. Inviscid drag rises from a numerically-near-zero level at $M_\infty = 0.15$ ($C_D = -0.0545$, a small residual-plateau artifact rather than a real negative drag, consistent with the modest +0.20-order residual reduction discussed in section 12) to $C_D = 0.191$ at $M_\infty = 0.8$ (transonic wave drag from the shock visible in `naca0012_m080_inviscid_mach_zoom.png`) and $C_D = 0.093$ at $M_\infty = 2.0$ (supersonic wave drag from the detached bow shock visible in `naca0012_m200_inviscid_mach_zoom.png` where the Mach contour shows the characteristic leading-edge shock stand-off and post-shock deceleration). Adding viscosity at $Re = 5000$ raises drag substantially at every Mach number

(e.g. $C_D = 0.760$ at $M_\infty = 0.15$ versus -0.055 inviscid), split into pressure and viscous contributions in Table 8; the viscous (skin-friction) fraction of total drag is largest at $M_\infty = 0.15$ ($0.142/0.760 \approx 19\%$) and smallest at $M_\infty = 2.0$ ($0.042/0.123 \approx 34\%$, still substantial despite the dominant wave-drag contribution at this Mach number).

Shock structure. The $M_\infty = 0.8$ Mach contour shows a near-uniform high-subsonic freestream ($M \approx 0.8$) everywhere except a localized, higher-Mach pocket near the leading edge consistent with a weak transonic shock; the $M_\infty = 2.0$ contour shows a clear detached bow shock ahead of the leading edge (a blue/low-Mach region immediately upstream of the airfoil nose, transitioning back to the yellow $M \approx 2$ freestream color a short distance away), exactly the qualitative structure expected for a blunt-nosed body in supersonic flow.

Viscous boundary layer. For the three Re 5000 laminar cases, every no-slip surface row in `surface.csv` reports `u=v=mach=0` exactly (section 5) while the adjacent skin-friction coefficient `cf` is nonzero and varies smoothly along the chord (`*_surface.cf.png`), consistent with a resolved (not wall-modeled) viscous boundary layer at this mesh resolution.

9.3 Cylinder Reynolds 20

`cylinder_m010_laminar_re20` is reported **failed**. A full 30,000-step production run (the case’s specified `max_steps`, unmodified) with the current `jac_scale=1024` default reaches only `residual_reduction_order=2` (far short of the 5-order target), and an automated tail-window-range heuristic in the solver driver initially mislabeled this run `statistically_periodic` because the last 3000 steps’ $\|R\|_2$ happened to fall in a bounded numeric range (39.3–42.1). This was caught and corrected by a dedicated audit: (1) restarting from the same final state and continuing for 11,000 further pseudo-time steps under identical real numerics shows the residual climbing *monotonically and with accelerating slope* the entire time (every one of the 3000 samples in the original run’s own tail window is already strictly increasing step-over-step; $\|R\|_2$ reaches 91.2 by step 40,000, still accelerating); and (2) the physical drag coefficient in `forces.csv` never stabilizes anywhere in the trajectory — it decays continuously from an impulsive-start value near $C_D = 268$, crosses zero at step 25,172, and continues into unphysically growing negative territory ($C_D = -1.23$ at the final step), which is decisive independent evidence that no genuine steady wake was ever reached, including during the window the automated heuristic flagged as a plateau. A `CFD2D_DEBUG_ARGMAX` trace on the restart extension shows the worst-residual cell is stationary at a far-field boundary cell (radius ≈ 10.4 from the cylinder, near the mesh’s outer boundary), distinct from the earlier `jac_scale=128` run’s much faster near-wall traveling-wave divergence (NaN by step 4852, preserved for reference in `results/mech2_final_cyl_re20_full/`) and consistent with the same broad LU-SGS-Jacobian-under-representation family discussed as Finding 4 in section 12, generalized here to the outer/far-field mesh region rather than a mid-chord or trailing-edge cell cluster. Residual-history, force-history, wall C_p/C_f , and Mach/velocity- magnitude/vorticity contours for the reported run are in `report/figures/cylinder_m010_laminar_re20_*.png`; the near-body velocity-magnitude and vorticity zooms still show a physically recognizable (if not fully converged) recirculating wake structure immediately behind the cylinder, though the far-field residual growth means this cannot be presented as a converged final state.

9.4 Cylinder Reynolds 200 Vortex Street

`cylinder_m010_laminar_re200` is reported **failed**. This is a genuine, honestly documented partial run: only 74 of the required 30,000 physical steps ($t = 0.74$ of the required $t_f = 300.0$) were completed, using the case’s fully unmodified production settings (`time_step=0.01`, `max_inner_iterations=1000`,

`inner_residual_reduction_target=10-3`). The root cause is not a bug but a measured, structural feasibility gap: every one of the 74 completed physical steps exhausted the full 1000-iteration BDF2 inner budget without reaching the target (section 6’s inner-solve statistics), because the `jac_scale=1024` value needed for stability through the wake-onset region caps the outer iteration’s convergence rate to near-linear ($\sim 0.4\text{--}4\%$ residual reduction per inner iteration). A dedicated feasibility investigation this session confirmed there is no simple fix: lower `jac_scale` values converge faster in isolation but reintroduce genuine divergence once real wake dynamics develop (`jac_scale=16` diverges by step 6–10, even faster than the original `jac_scale=128`’s step-56 divergence; `jac_scale=256` survives the full known danger window but still only reaches a best inner-residual ratio of $\approx 3.6 \times 10^{-3}$, short of the 10^{-3} target), and a reactive/adaptive `jac_scale` scheme (mirroring the existing CFL-safety-cutback design) also diverged, even earlier than the fixed value. Real MPI scaling was measured directly on this case: $np=4$ gives a genuine $4.63\times$ per-step speedup (matching residuals across rank counts to 5+ significant figures), which reduces the corrected full-run wall-clock estimate from ≈ 1138 hours ($np=1$) to ≈ 246 hours (≈ 10.2 days, $np=4$) — MPI parallelism helps substantially but does not close the feasibility gap within a realistic session.

No Strouhal number or shedding-frequency estimate is reported. Seventy-four physical-time units captures at most 1–1.5 partial oscillation cycles at the expected shedding period ($St \approx 0.19\text{--}0.20 \Rightarrow \text{period} \approx 5$ time units), far short of what any genuine spectral (FFT) or zero-crossing frequency estimate requires; an FFT run on this short window returns a numerically well-defined but physically meaningless value (dominant frequency ≈ 5.4 , an aliased artifact of the short window, not a real shedding frequency), and is deliberately not reported as a Strouhal number in Table 8 or anywhere else in this report. The completed 74-step force history does show the very earliest hint of a developing lift oscillation (peak-to-peak C_L amplitude ≈ 0.026 over the final third of the run, per `report/sanity_checks.json`’s `re200_nonzero_lift_variation` check), i.e. the wake has begun to break its initial top/bottom symmetry as physically expected, but not nearly enough of the transient has developed to call this a statistically periodic vortex street. The near-body vorticity and velocity-magnitude contours (`report/figures/cylinder_m010_laminar_re200_vorticity_zoom.png`, `*_velmag_zoom.png` vorticity clipped to $[-5, 5]$ per `NUMERICAL_PARAMETERS.md`) show this early asymmetry directly: a thin band of concentrated vorticity forming on each side of the cylinder with a visibly asymmetric, just-beginning-to-tilt wake, consistent with shedding onset rather than an established periodic street.

10 Parallel Validation

One NACA case (`naca0012_m080_inviscid`) and one cylinder case (`cylinder_m010_laminar_re20`) were each run at $np = 1, 2, 4, 8$ using an identical case JSON and numerics configuration, to demonstrate the MPI decomposition is producing the same physics regardless of rank count (TASK.md disqualification-trigger check). These parallel-validation runs used `jac_scale=128` (the value in effect earlier in this project’s development), since their purpose is demonstrating rank-count invariance of a fixed configuration, not reproducing this report’s final case results (section 9, which use the later `jac_scale=1024` default); rank-count invariance itself does not depend on which `jac_scale` value is used.

Bit-identical first step. For both cases, step 1 of `residuals.csv` is byte-identical across all four rank counts (e.g. `naca0012_m080_inviscid`: $\|R\|_2 = 0.065827$ at every np ; `cylinder_m010_laminar_re20`: $\|R\|_2 = 58.633$ at every np), confirming the very first residual assembly — including the halo exchange, boundary treatment, and reduction across ranks — is numerically identical regardless of how the mesh is partitioned.

Trajectory tracking. Both cases’ full trajectories track closely across rank counts throughout the run, not just at step 1. For `naca0012_m080_inviscid` at step 8000 (the common point available across all four np logs), C_D is 0.14536 ($np=1$), 0.14557 ($np=2$), 0.14564 ($np=4$), 0.14587 ($np=8$) — a 0.35% spread, consistent with small floating-point-order-of-summation differences in the MPI reduction rather than a genuinely different converged state. For `cylinder_m010_laminar_re20` at step 100, $\|R\|_2$ is 17.663 ($np=1, 2$, identical) and 17.677–17.680 ($np=4, 8$), a $< 0.1\%$ spread; by step 3000, all four rank counts remain within $\approx 1\%$ of each other’s $\|R\|_2$, well before this case’s later divergence (section 9.3) makes further comparison meaningless.

Load balance. Tables 5 and 6 (repeated from section 7) show owned-cell counts balanced to within 1% of the mean at $np = 8$ for both meshes, with edge cuts and ghost-cell counts typical of a well-formed METIS k -way partition; no rank is left significantly over- or under-loaded at any tested rank count.

Scaling. Both cases show the expected strong-scaling pattern: substantial speedup from $np = 1 \rightarrow 4$ as the per-rank cell count drops roughly in proportion, with diminishing returns from $np = 4 \rightarrow 8$ as communication overhead (more neighbor ranks per partition, smaller per-rank work) becomes a larger fraction of the shrinking per-rank compute time — the qualitatively expected behavior for a $\sim 20k$ – $10k$ -cell mesh at these rank counts, not evidence of an MPI correctness problem.

11 Visualization and Plot Style Requirements

All line and field plots for this report are produced by the plotting scripts under `solver/tools/` directly from submitted CSV/field files, using Matplotlib for line plots and Matplotlib triangulation or PyVista for unstructured field contours. Every field visualization uses a labeled colorbar with the plotted variable name and units/nondimensional label; every line plot uses labeled axes, a legend, grid/tick marks, and nondimensional quantity names, with consistent color choices for the same quantity/case across figures. Both Mach and pressure contours are produced for every case, with file names matching the plotted variable (e.g. `*_mach.png` always plots Mach number, `*_pressure.png` always plots pressure), and near-body/near-wake zooms are used in addition to whole-domain views so that boundary-layer, shock, and wake structure remain visible. All 69 figures actually produced for this report are listed in `report/figure_manifest.csv`, which maps every figure file to its case id, figure type, plotted variable, and source data file; the manifest was validated by the benchmark’s own `examiner/validate_outputs.py` (`validate_report`), which confirms every filename-variable pair is consistent (e.g. every `*_mach.png` entry has `variable` containing `mach`) and that every case has at least one Mach and one pressure figure entry, plus a vorticity/velocity entry for both cylinder cases. Each NACA case contributes a residuals, forces, surface- C_p plot, and (laminar cases only) surface- C_f plot, plus whole-domain and near-body-zoom Mach and pressure contours; each cylinder case additionally contributes whole-domain and near-wake-zoom vorticity and velocity-magnitude contours.

12 Limitations and Failure Analysis

Two of the eight required cases are failures. Per each case’s own `run_status.json`: `cylinder_m010_laminar`’s `convergence_status` field is `"failed"` (an accelerating, non-periodic residual runaway with no genuine stable plateau, section 9.3), and `cylinder_m010_laminar_re200`’s `convergence_status` field is likewise `"failed"` (a genuine 74-of-30000-step partial run, halted by a documented infeasible

per-step compute cost, section 9.4). Neither case is described as converged or statistically periodic anywhere else in this report.

Finding 4: mid-chord LU-SGS Jacobian under-representation. The flagship `naca0012_m015_inviscid` case converges to only +0.20 orders of residual reduction (target: 4 orders), plateauing rather than continuing to decay. After five dedicated investigation sessions, the root cause was traced to a mid-chord, aspect-ratio-graded cell cluster (around $x \approx 0.674$) where the local face-area-to-cell-thickness ratio reaches $\sim 450\times$ the typical value, and the LU-SGS scheme’s scalar (not full-block) Jacobian under-represents the true local face stiffness by enough to force the CFL-safety mechanism into a continuous, deep cutback (`cfl_safety_cutback_events=12365` out of 20,000 steps for this case, `final_cfl_safety_scale` ≈ 0.003). Exhaustively ruled out as contributing causes across this project’s history: limiter choice, least-squares gradient pruning/weighting, Roe entropy-fix parameters (tested up to floor fractions of 2.0, byte-identical to baseline), gradient-magnitude capping, `dt_smoothing_passes` (30/100/300, made it worse), `max_inner_iterations` (50/300, made it worse), and a direct ablation of the CFL-safety mechanism (which proved the cutback is correctly reacting to real residual growth, not falsely triggering). This remains an accepted, documented limitation: the root cause (Jacobian under-representation of extreme local stiffness) is identified but not fixed. **The same broad mechanism family, generalized to different mesh regions**, is implicated in both of this project’s failed cases: a far-field/outer-boundary manifestation in `cylinder_m010_laminar_re20` (section 9.3), and the wake-onset viscous instability that forces `cylinder_m010_laminar_re200`’s large `jac_scale` (section 9.4), which in turn caps that case’s achievable inner-loop convergence rate.

Trailing-edge-cusp late-onset divergence (found and fixed). A second, distinct mechanism was found this project: a tiny-volume ($\sim 10^{-9}$) trailing-edge cusp cell cluster on the NACA mesh ($x \approx 1.0049\text{--}1.0053$, where the airfoil’s upper and lower surfaces pinch together) caused a divergence that only manifested deep into full-length production runs, past previously-validated shorter debug windows, in five of the six NACA cases plus both cylinder cases. Raising the LU-SGS diagonal scale `jac_scale` from 128 to 1024 (`src/solver/context.hpp`) genuinely fixed this for `naca0012_m200_inviscid` (previously NaN at step 30435, now a clean full 40000-step run) and `naca0012_m015_laminar_re5000` (previously -11.78 orders, now $+1.12$ orders full-length), and incidentally stabilized `naca0012_m080_inviscid`’s unrelated slow-drift instability as a side effect. For `naca0012_m080_laminar_re5000` and `naca0012_m200_laminar_re5000`, the fix only *delays* rather than eliminates the same mechanism for viscous flow (it resumes at a later step); both cases use a documented earlier stop on a verified stable plateau with margin before the later recurrence, per `NUMERICAL_PARAMETERS.md`’s explicit allowance. This same fix was also required for `cylinder_m010_laminar_re200` to survive its wake-onset region at all (without it, that case diverges catastrophically by $t = 0.56\text{s}$), but the large `jac_scale` this requires is precisely what makes that case’s inner BDF2 loop too slow to converge within its iteration budget (section 9.4) — a direct robustness/convergence-rate tradeoff inherent to a scalar-diagonal LU-SGS scheme, not a bug.

MPI/parallel behavior. No MPI load imbalance or communication problem was observed: partition diagnostics (tables 5 and 6) show owned-cell counts balanced to within 1% of the mean at $np = 8$ for both tested meshes, and section 10 shows bit-identical first-step residuals and closely-tracked trajectories across $np = 1, 2, 4, 8$ for both a NACA and a cylinder case.

Positivity and shock robustness. The positivity fallback was never triggered in any of the six converged/statistically-periodic cases, including both supersonic $M_\infty = 2.0$ NACA cases, indicating the Roe/Harten–Hyman flux and limiter combination resolves the detached bow shock without requiring a positivity rescue at this mesh resolution.

What more compute/time would improve. (1) A full-block (not scalar) LU-SGS Jacobian,

or a matrix-free Newton–Krylov inner solver, would likely resolve both Finding 4 and the trailing-edge-cusp/wake-onset family directly, rather than requiring an over-relaxed diagonal scale that trades convergence rate for robustness. (2) A finer, more isotropic mesh in the affected regions (mid-chord for Finding 4, trailing-edge cusp, and the cylinder far-field/outer boundary) would reduce the extreme local face-stiffness ratios driving both mechanisms. (3) For `cylinder_m010_laminar_re200` specifically, a genuinely faster inner solver (rather than more compute time alone, given the measured ≈ 10 -day $np=4$ estimate) would be required to reach the full 30,000-step target and obtain enough shedding periods for a real Strouhal estimate; alternatively, a longer session budget combined with the measured $4.63 \times np=4$ speedup could still complete it given enough wall-clock time. (4) For `cylinder_m010_laminar_re20`, since no `jac_scale` value tested this session produces a stable plateau, the same full-block Jacobian or Newton–Krylov improvement in (1) is the most promising path, rather than further scalar-`jac_scale` tuning.